

Tableau Lab 02

Interactive Profitability & Customer Analysis – Sample Superstore

TURN DATA
INTO INSIGHTS

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Scenario

Management understands the overall sales picture, but now wants to know why profitability changes. Build an interactive analysis that lets the user switch metrics, identify loss areas, investigate customers, and focus on the most important records.



Learning Goals

- Create and use calculated fields for business analysis.
- Use table calculations, parameters, and sets/Top N logic.
- Build interactions with dashboard actions rather than relying only on visible filters.
- Design one advanced dashboard that supports investigation and decision-making.

1 Part 1 – Create Calculated Fields

View	Business requirement
Profit Margin	$\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$. Format as a percentage.
Profit Status	Classify records as Profit or Loss based on the Profit value.
Shipping Days	Calculate the number of days between Order Date and Ship Date using DATEDIFF .

Use the calculated fields in at least one visualization each.

2 Part 2 – Table Calculation Practice



- Create a view showing Sales by Region and display **Percent of Total** so the user can see each Region's contribution.
- Create a monthly Sales view and add a **Running Total**. Confirm that the calculation is computing in the correct direction.

3 Part 3 – Dynamic Metric Parameter



- Create a parameter named **Metric Selector** with the values: Sales, Profit, Quantity.
- Create a calculated field named **Selected Metric** that returns the measure selected by the parameter.
- Build a Sub-Category chart using Selected Metric so the same chart changes when the parameter changes.
- Make the chart title or nearby text clearly indicate which metric the user is viewing.

4 Part 4 – Dynamic Top N



- Create an integer parameter called **Top N** (recommended range: 5 to 20).
- Create a Customer set based on the Top N customers by Sales.
- Build a sorted bar chart that responds to the Top N parameter and shows Sales and Profit in the tooltip.

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5 Part 5 – Build the Investigation Views

View	Business requirement
Customer Scatter Plot	Plot Sales vs Profit by Customer Name. Use Profit Status to make loss-making customers easy to identify. Include useful details in the tooltip.
Profitability Heatmap	Create a heatmap using Region and Sub-Category . Color the view by Profit Margin so weak combinations stand out.
Shipping Analysis	Compare average Shipping Days by Ship Mode . Keep the view simple and clearly formatted.
Dynamic Metric View	Use the Metric Selector chart created earlier as one of the main analytical views.
Top Customers	Include the dynamic Top N customer view so management can focus on the most commercially important customers.

6 Part 6 – Dashboard Actions



- Create **one advanced dashboard** that contains the most useful views from this lab.
- Add at least one **Filter Action**: selecting a mark in one chart must filter another chart.
- Add at least one **Highlight Action** or another meaningful interaction.
- Keep the Metric Selector and Top N parameters visible and easy to use.
- Make sure users can clear a selection and return to the full dashboard.

7 Part 7 – Business Questions



1. Which Region/Sub-Category combinations have the weakest Profit Margin?
2. Can you identify a high-sales customer who is generating a loss?
3. Which Ship Mode has the longest average Shipping Days?
4. How does the ranking of Sub-Categories change when you switch between Sales and Profit?
5. What management action would you recommend based on your dashboard?

Design & Submission

- Prioritize investigation: the dashboard should help a user move from a summary to a specific problem.
- Do not overcrowd the page. It is better to use four or five useful views than many small charts.
- Use consistent formatting for currency, percentages, titles, labels, and tooltips.
- Submit the Tableau workbook (.twb or .twbx), an exported dashboard image/PDF, and the five short business answers.

Optional Bonus



Use a **Context Filter** to demonstrate how Tableau's filter order can change a Top N result, or add a navigation button to a second detail dashboard. The core assignment requires only one strong dashboard.